RACE Regular Analysis Through Continuous Exercise

Sequences and Series : RACE-01

Single Correct Question

1. Find sum $\frac{3}{4},\frac{2}{3},\frac{7}{12}$,..... to 19 terms.

(A) 0 (B) 1 (C) –1 (D) 2

MSS156

2. Find sum 1.3,–3.1, –7.5,.... to 10 terms

(A) –184 (B) –185 (C) –186 (D) –187

MSS157

3. Find the sum of first 24 terms of the A.P. a_1, a_2, a_3....., if it is known that $a_1 + a_5 + a_{10} + a_{15} + a_{20} + a_{24} = 225$.

(A) 800 (B) 825 (C) 850 (D) 900

MSS158

4. In an A.P. the first term is 2, the last term is 29, the sum is 155; find the common difference.

(A) 1 (B) 2 (C) 3 (D) 4

MSS159

5. The third term of an A.P. is 18, and the seventh term is 30; find the sum of 17 terms.

(A) 611 (B) 612 (C) 613 (D) 614

MSS160

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Subjective Type Question

6. Write down the sequence whose n^{th} terms is :

(a) $\frac{2^n}{n}$ **MSS161**

(b) $\frac{3+(-1)^n}{3^n}$ **MSS162**

7. For an A.P, show that $t_m + t_{2n+m} = 2t_{m+n}$

MSS163

8. In an A.P. of which 'a' is the Ist term, if the sum of the Ist p terms is equal to zero, show that the sum of the next q terms is $-\left(\frac{aq(p+q)}{p-1}\right)$ **MSS164**

9. The sum of three numbers in A.P. is 12, and the sum of their cubes is 408; find them.

MSS165

10. In an A.P. if the sum of 7 terms is 49, and the sum of 17 terms is 289, find the sum of n terms.

MSS166

ANSWER KEY

Question	1	2	3	4	5
Answer	A	B	D	C	B

6. (a) $\frac{2}{1},\frac{4}{2},\frac{8}{3},\frac{16}{4}$,..... (b) $\frac{2}{3},\frac{4}{9},\frac{2}{27},\frac{4}{81}$,..... **9.** (1,4,7) **10.** (n^2)

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Regular Analysis Through Continuous Exercise

Sequences and Series : RACE-02

Single Correct Question

1. If the sum of n terms of an A.P. is $2n + 3n^2$, find the rth term.
(A) $6r + 1$ (B) $6r - 1$ (C) $6r + 3$ (D) $6r - 3$

MSS167

2. Find a three-digit number whose consecutive digits form a G.P. If we subtract 792 from this number, we get a number consisting of the same digits written in the reverse order. Now, if we increase the second digit of the required number by 2, then the resulting digits will form an A.P.
(A) 931 (B) 842 (C) 421 (D) None of these

MSS168

3. If the third term of G.P. is 4, then find the product of first five terms.
(A) 4^4 (B) 4^5 (C) 2^7 (D) 2^9

MSS169

4. Find the sum of the series $\frac{3}{4}, 1\frac{1}{2}, 3, \ldots$ to 8 terms.
(A) 191 (B) $191\frac{1}{2}$ (C) $191\frac{1}{4}$ (D) None of these

MSS170

5. Find the sum of the series $\frac{8}{5}, -1, \frac{5}{8}, \ldots$ upto infinity
(A) $\frac{64}{65}$ (B) $\frac{63}{65}$ (C) $\frac{65}{63}$ (D) $\frac{65}{64}$

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MSS171

6. Find the sum of the series .45, .015, .0005, .. upto infinity
(A) $\frac{29}{50}$ (B) $\frac{27}{58}$ (C) $\frac{1}{2}$ (D) None of these

MSS172

7. The sum of the first 6 terms of a G.P. is 9 times the sum of the first 3 terms; find the common ratio.
(A) 2 (B) 3 (C) 4 (D) None of these

MSS173

Subjective Type Question

8. The sum of n terms of two arithmetic series are in the ratio of $(7n + 1) : (4n + 27)$. Find the ratio of their n^{th} term.

MSS174

9. Find three numbers in G.P., whose sum is 52 and the sum of whose products in pairs is 624.

MSS175

10. The continued product of three numbers in G.P. is 216, and the sum of the product of them in pairs is 156; find the numbers.

MSS176

ANSWER KEY

Question	1	2	3	4	5	6	7
Answer	B	A	B	C	A	B	A

8. $(14n - 6)/(8n + 23)$ **9.** 4, 12, 36 **10.** 2,6,18

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RACE
Regular Analysis Through Continuous Exercise

Sequences and Series : RACE-03

Single Correct Question

1. In a GP the ratio of the sum of the first eleven terms to the sum of the last eleven terms is 1/8 and the ratio of the sum of all the terms without the first nine to the sum of all the terms without the last nine is 2. Find the number of terms in the GP.

(A) 35 (B) 36

(C) 37 (D) 38

MSS177

2. If the 7th term of a H.P. is 8 and the 8th term is 7. Then find the 28th term.

(A) 3 (B) 2 (C) 4 (D) $\frac{1}{2}$

MSS178

3. In a H.P., if 5th term is 6 and 3rd term is 10. Find the 2nd term.

(A) 15 (B) 12 (C) 14 (D) 16

MSS179

4. If the 10th term of a HP is 21 and 21st term of the same HP is 10, then find the 210th term.

(A) 1 (B) 2 (C) 3 (D) 4

MSS180

5. If $\frac{a^n+b^n}{a^{n-1}+b^{n-1}}$ is the G.M. between distinct positive real numbers a and b then find the value of 'n'.

(A) 2 (B) $\frac{1}{2}$

(C) 3 (D) $\frac{1}{3}$

MSS181

Subjective Type Question

6. The sum of an infinite number of terms of a G.P. is 4, and the sum of their cubes is 192; find the series.

MSS182

7. Sum of following series : $x + a, x^2 + 2a, x^3 + 3a$ ... to n terms.

MSS183

8. If the pth, q^{th} and r^{th} terms of a H.P. are a, b, c respectively, then prove that $\frac{q-r}{a}+\frac{r-p}{b}+\frac{p-q}{c}=0$.

MSS184

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9. Find the fourth term in each of the following Harmonic progressions :

(a) $2, 2\frac{1}{2}, 3\frac{1}{3}, \ldots$ **MSS185**

(b) $1\frac{1}{2}, 2, 3 \ldots$ **MSS186**

(c) $3\frac{3}{5}, 6, 18, \ldots$ **MSS187**

10. If a, b, c be in H.P., show that $a : a - b = a + c : a - c$. **MSS188**

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ANSWER KEY

Question	1	2	3	4	5
Answer	D	B	A	A	B

6. $6, -3, 1\frac{1}{2}, \ldots$ **7.** $\frac{x\left(x^n - 1\right)}{x - 1} + \frac{\left(n\left(n+1\right)\right)a}{2}$ **9.** (a) 5 (b) 6 (c) –18

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RACE
Regular Analysis Through Continuous Exercise

Sequences and Series : RACE-04

Single Correct Question

1. The harmonic mean of two numbers is 4. The arithmetic mean A & the geometric mean G satisfy the relation $2A + G^2 = 27$. Find the sum of two numbers.

(A) 7 (B) 8 (C) 9 (D) 10

MSS189

2. Find sum $1 + 2a + 3a^2 + 4a^3 + \cdots$ to n terms.

(A) $\frac{1-a^n}{(1-a)^2} - \frac{na^n}{1-a}$ (B) $\frac{1-a^n}{(1-a)^2} + \frac{na^n}{1-a}$

(C) $\frac{1+a^n}{(1+a)^2} - \frac{na^n}{1+a}$ (D) $\frac{1+a^n}{(1+a)^2} + \frac{na^n}{1+a}$

MSS190

3. Find sum $1+\frac{3}{4}+\frac{7}{16}+\frac{15}{64}+\frac{31}{256}+\ldots$ to infinity.

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(A) $\frac{5}{3}$ (B) $\frac{7}{3}$ (C) $\frac{8}{3}$ (D) $\frac{10}{3}$

MSS191

4. Find sum $\frac{2}{3}+\frac{3}{3^2}+\frac{2}{3^3}+\frac{3}{3^4}+\frac{2}{3^5}+\frac{3}{3^6}+\ldots$ to infinity.

(A) $\frac{9}{8}$ (B) $\frac{9}{2}$ (C) $\frac{7}{8}$ (D) $\frac{7}{2}$

MSS192

Subjective Type Question

5. If b is the harmonic mean between a and c, then prove that $\frac{1}{b-a}+\frac{1}{b-c}=\frac{1}{a}+\frac{1}{c}$.

MSS193

6. Insert 19 arithmetic means between $\frac{1}{4}$ and $-9\frac{3}{4}$.

MSS194

7. Insert 5 geometric means between $3\frac{5}{9}$ and $40\frac{1}{2}$.

MSS195

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8. If between any two quantities there be inserted two arithmetic means A_1, A_2; two geometric means G_1, G_2 ; and two harmonic means H_1, H_2; show that $G_1G_2 : H_1H_2 = (A_1 + A_2) : (H_1 + H_2)$.

MSS196

9. Given that the cubic $ax^3 - ax^2 + 9bx - b = 0$ $(a \neq 0)$ has all three positive roots. Find the harmonic mean of the roots independent of a and b, hence deduce that the root are all equal. Find also the minimum value of $(a + b)$ if a and $b \in N$.

MSS197

10. Find the sum of n terms of the series the rth term of which is $(2r + 1)2^r$.

MSS198

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ANSWER KEY

Question	1	2	3	4
Answer	C	A	C	A

6. $-\frac{1}{4}, -\frac{3}{4}, \ldots, -9\frac{1}{4}$ **7.** $\frac{16}{3}, 8, \ldots, 27$ **9.** (28) **10.** $n.2^{n+2} - 2^{n+1} + 2$

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RACE
Regular Analysis Through Continuous Exercise

Sequences and Series : RACE-05

Single Correct Question

1. Find the sum of the infinite series $\frac{1.3}{2}+\frac{3.5}{2^2}+\frac{5.7}{2^3}+\frac{7.9}{2^4}+.....\infty$.

(A) 27 (B) 25

(C) 23 (D) 21

MSS199

2. Find the sum of the series upto n terms $1+\frac{1+2}{2}+\frac{1+2+3}{3}+\frac{1+2+3+4}{4}+.....$

(A) $\frac{n(n+3)}{4}$ (B) $\frac{n(n+2)}{4}$

(C) $\frac{n(n+3)}{2}$ (D) $\frac{n(n+2)}{2}$

MSS200

3. The value of $1^2 + 3^2 + 5^2 + ... + 25^2$ is :

(A) 1728 (B) 1456

(C) 2925 (D) 1469

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MSS201

4. The sum of the series : $(2)^2 + 2(4)^2 + 3(6)^2 + \cdots$ upto 10 terms is :

(A) 11300 (B) 12100

(C) 12300 (D) 11200

MSS202

5. The value of $(1.1^2 + 3.2^2 + 5.3^2 + ... +$ upto 10 terms) is equal to :

(A) 6050 (B) 5965

(C) 5665 (D) 5385

MSS203

Subjective Type Question

6. Find the sum of 'n' terms of the series whose n^{th} term is

(a) $3n^2 - n$. **MSS204**

(b) $3^n - 2^n$ **MSS205**

7. Sum the following series to n terms :

(a) $1 . 4 . 7 + 2 . 5 . 8 + 3 . 6 . 9 + ...$ **MSS206**

(b) $1 . 5 . 9 + 2 . 6 . 10 + 3 . 7 . 11 + ...$ **MSS207**

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8. Find the nth term and the sum of n terms of the series :

(a) 4, 14, 30, 52, 80, 114, ... **MSS208**

(b) 9, 16, 29, 54, 103, ... **MSS209**

9. Find the nth term and the sum to n terms of the sequence:

(i) 1 + 5 + 13 + 29 + 61 +..... **MSS211**

(ii) 6 + 13 + 22 + 33 +..... **MSS212**

10. Sum of the following series to n terms and to infinity (where it is finite and defined) :

(i) $\frac{1}{1.4.7}+\frac{1}{4.7.10}+\frac{1}{7.10.13}+.....$ **MSS213**

(ii) $\sum_{r=1}^{n}\frac{1}{4r^2-1}$ **MSS214**

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ANSWER KEY

Question	1	2	3	4	5
Answer	C	A	C	B	C

6. (a) $n^2(n+1)$ (b) $\frac{1}{2}\left(3^{n+1}+1\right)-2^{n+1}$

7. (a) $\frac{n}{4}(n+1)(n+6)(n+7)$ (b) $\frac{n}{4}(n+1)(n+8)(n+9)$

8. (a) $3n^2+n;\ n(n+1)^2$ (b) $3\,.\,2^n+n+2;\ 6(2^n-1)+\frac{n(n+5)}{2}$

9. (i) $2^{n+1}-3;\ 2^{n+2}-4-3n$ (ii) $n^2+4n+1;\ (1/6)\,n\,(n+1)\,(2n+13)+n$

10. (i) $s_n = (1/24) - [1/\{6(3n+1)\,(3n+4)\}]$; $s_\infty = 1/24$

(ii) $n/(2n+1)$; $s_\infty = 1/2$

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